

Vibeathon 6.0 (Vibecoding Hackathon) - July 2026



NXTGENSEC
SECURING DIGITAL ASSETS



TableFlow

AI-Powered Smart Restaurant Management System

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Hackathon: VibeAthon 6.0 – Professional Category



CURRENT PROBLEM

Restaurant Operations Are Still Manual Challenges

Manual table allocation

Long customer waiting time

Reservation conflicts

Billing inefficiencies

No AI-powered insights

Existing Solutions

Focus mainly on food delivery

Limited support for dine-in operations

Impact

Slow service

Poor customer experience

Reduced operational efficiency



Proposed Solution

TableFlow – One Platform for Complete Restaurant Operations

End-to-End Workflow

Reservation / Walk-in → Live Menu → Order → Table Allocation → Billing → Analytics

Key Features

Real-Time Menu Availability

Digital Order Placement

Smart Table & Waitlist Management

AI Menu Intelligence

Transparent Itemized Billing

Reservation Approval & Verification

Value

- ✓ Faster Operations
- ✓ Better Customer Experience
- ✓ Smarter Business Decisions



TECHNICAL APPROACH

Modern Full-Stack Architecture

Frontend

Next.js 14

TypeScript

Tailwind CSS

Backend

Supabase

PostgreSQL

Realtime

Google OAuth

AI & Cloud

Google Gemini 2.5 Flash

Resend Email

Vercel Deployment

Architecture

Next.js ⇌ Supabase ⇌ PostgreSQL ⇌ Gemini AI



AI & Smart Features

AI-Powered Restaurant Intelligence

Google Gemini AI

- Demand Forecasting
- Star & Deadweight Dish Analysis
- Customer Feedback Summary

Smart Automation

- Automatic Table Allocation
- Seat-Level Capacity Management
- Reservation Verification
- Real-Time Menu Updates



Use Cases & Impact

Real-World Benefits

Target Users

Restaurants

Cafés

Restaurant Owners

Managers

Business Impact

Faster Service

Better Table Utilization

Reduced Manual Work

Improved Customer Experience

Data-Driven Decision Making



Future Scope & Conclusion

Future Enhancements

Multi-Restaurant Support

QR Table Ordering

Kitchen Display System (KDS)

Inventory Management

Loyalty & Rewards

Payment Gateway Integration

Mobile Application

Scalable SaaS Architecture

Designed for future expansion across multiple restaurants

Conclusion

Thank You
TableFlow
AI-Powered Smart Restaurant
Management System
Live Demo
Website

<https://table-flow-nu.vercel.app>

GitHub

<https://github.com/ayush22cp008/TableFlow>
[W](#)

Team

Ayush008

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